

claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_3d536535-241\agent-aa77a48d6cd7021f4.jsonl`  
- SHA-256: `66c40e62de9551fdd94f03ad991c6d973552b439d8d383233039130530e9b1ce`  
- Source modified: `2026-06-16T03:24:15+00:00`  
- Imported at: `2026-07-05T16:48:27+00:00`  
- project: `wf_3d536535-241`  
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`
```

Transcript

1. user (2026-06-16T03:23:06.747Z)

Adversarial Claim Verifier (voter 3/3)

Be SKEPTICAL. Try to REFUTE this claim. 2/3 refutations kill it.

Research question

Research question: **What concrete, recent (2022-2025) techniques and evaluation practices should a local, commercially-licensed badminton rally-detector adopt to close the quality gap with a strong cloud VLM (Gemini), given a tiny labeled set (~6 - ~16 golden videos)?** Produce an adversarially-verified, cited report organized around the four dimensions below. Prefer specific named methods, papers (arXiv/venue), and reported numbers over generic advice; flag anything you cannot verify.

CONTEXT (so you target the real gap, not generic ML):

- System: a local pipeline detects the shuttle per-frame (WASB / TrackNet, MIT-licensed) ? a heuristic turns the trajectory into candidate rally windows ? today an optional Gemini "handoff" confirms rallies. We just measured a fully-local (no-AI) run vs Gemini on 3 matches: local emits far FEWER but MUCH LONGER windows (mean 65s vs 6.1s; one window spanned a whole 174s clip) ? it does NOT miss activity, it FAILS TO SEGMENT. Root cause (code-confirmed): a frame is "active" iff the shuttle is visible AND moving above a small per-second-normalized velocity threshold; active frames within a 2 s gap are merged; there is NO max-window cap and NO inactivity/boundary split. So "shuttle is moving" is conflated with "rally in progress," and dead-time + serves + knock-ups bridge rallies into mega-windows. Shuttle tracking itself is excellent (96-99% per-frame visibility).

- HARD CONSTRAINTS: weights/data/code must be MIT/Apache-2.0/BSD only (no viral/NC/custom licenses, reject Gemma-3/Llama-4-MAU-cap); we may NEVER train owned weights on paid-LLM (Gemini/OpenAI/Anthropic) outputs (terms/copyright), though paid LLMs may be used at inference; minors excluded from any training pool; runs on a single local GPU.

- WHAT WE ALREADY HAVE (do NOT re-survey the basics ? go deeper/newer): the architecture skeleton is known to be standard Temporal Action Localization/Segmentation (proposal+scorer) + late/score-level fusion + stacked generalization; in-domain prior art (MonoTrack, ShuttleSet, See et al. 2024/25 non-broadcast badminton rally start/end, Ghosh et al. point-segmentation, tennis/TT play-break state machines) is already cited. We have a working eval harness: leave-one-video-out (grouped by match), temporal-IoU F1 @0.5, F1@{0.1,0.25,0.5}, mAP@tIoU, segment-count-ratio, bootstrap 95% CIs, paired exact sign-test, Platt/isotonic calibration + Brier/ECE, a tiny numpy LogisticRegression scorer over per-window feature columns, an experiment leaderboard, and an "eval+serve" guard (a cue measured +0.074 F1 on permissive proposal windowing but ?0.008 on the coarser served windowing ? reverted). Available but default-OFF per-window signals: net-crossing count (rally gate), 5 person/court features, 3 audio-hit features.

THE FOUR DIMENSIONS:

1) WINDOWING & BOUNDARIES (Temporal Action Localization/Segmentation): beyond proposal+score ? what are the best **recent** methods for (a) turning a dense per-frame signal into well-segmented

actions, (b) explicitly controlling OVER-segmentation/merging, (c) boundary detection/regression and splitting merged segments? Cover e.g. MS-TCN/MS-TCN++, ASRF (action-boundary regression), BMN/BSN boundary-matching, ActionFormer/TriDet/TemporalMaxer (actionness + boundary heads), smoothing/boundary losses, and the standard metrics for over-segmentation (segmental F1@k, edit/Levenshtein score, mAP@tIoU). Which transfer to a 1-D shuttle-trajectory + auxiliary-cue setting on tiny data?

2) SHUTTLE-TRAJECTORY ? RALLY SIGNAL PROCESSING (racket-sports specific): how do published systems convert ball/shuttle tracking into rally/point segments and distinguish "in-play" from dead-time? Cover rally-state machines, serve/let detection, net-crossing & two-sided-occupancy cues, hit/contact (audio + kinematic) detection, and fps-invariant trajectory features. What features best separate "rally" from "shuttle merely moving"?

3) SMALL-DATA STRATEGIES: with ~6?16 labeled videos, what gives the most quality per label ? active learning (which videos/segments to label next; uncertainty/diversity/core-set), semi-/weakly-/point-supervised TAL (e.g. SF-Net, timestamp supervision), self-training/pseudo-labeling, and data-efficient TAL? Separately: the legally-clean ways to exploit a strong teacher (Gemini) at INFERENCE for weak/pseudo-labels or as an evaluation oracle WITHOUT its outputs becoming training data for owned weights ? what does the literature say about teacher-student / distillation boundaries and label-noise handling, and what governance distinctions matter?

4) EVALUATION METHODOLOGY & RIGOR at small n: trustworthy measurement with ~6?16 videos ? bootstrap/permutation tests, paired sign-tests, leave-one-group-out vs RL00CV bias, calibration (reliability, ECE), the over-segmentation-blindness of plain tIoU, shadow/A-B evaluation, regression gates, and concrete pitfalls (label noise, annotator agreement/IAA, multiple-comparison/over-fitting traps, eval-vs-serve drift). What metrics + protocols should a small-data sports-segmentation leaderboard standardize on?

Deliverable: a structured, cited report with, per dimension, the 3?5 highest-leverage techniques that APPLY to our constrained setting (small data, local, license-clean, 1-D trajectory + cues + learned scorer), each with: what it is, why it fits our over-merge/small-data problem, the expected difficulty/payoff, and the citation. Call out which ideas are low-effort high-confidence wins vs research bets. Be explicit about anything unverified or license-incompatible.

Claim under review

"Timestamp supervision (action labels for only a sparse set of frames, e.g. one frame per action, then iteratively refine pseudo frame-wise labels) achieves performance comparable to fully supervised methods, making it a compelling low-label direction."

****Source:**** <https://arxiv.org/pdf/2210.10352> (primary)

****Supporting quote:**** "The general strategy of timestamps is to generate pseudo frame-wise labels for supervision, then refine them iteratively. Notably, timestamp methods perform comparably with fully supervised approaches, making it a compelling direction for further exploration."

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-06-16T03:23:09.332Z)

I'll verify this claim about timestamp supervision. Let me examine the primary source and search for contradicting evidence.

3. assistant (2026-06-16T03:23:12.905Z)

Let me fetch the primary source to check whether the quote actually supports the claim.

4. user (2026-06-16T03:23:18.875Z)

I apologize, but I cannot provide a useful response to your question. The document you've provided appears to be a PDF file encoded in binary/compressed format (showing raw PDF stream data with FlateDecode compression). The content is not in readable text form.

To answer your questions about timestamp supervision for temporal action segmentation, I would need:

1. A readable text version of the paper
2. The paper's title, authors, and venue information
3. The actual content sections discussing timestamp supervision methods

If you could provide:

- The paper as plain text
- A link to the paper
- The paper's title so I can identify it
- Relevant excerpts in readable format

Then I could accurately quote the paper's claims about timestamp supervision performance, any caveats mentioned, and the specific wording used regarding this methodology.

[Binary content (application/pdf, 1.8MB) also saved to C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\tool-results\webfetch-1781579999760-41alwg.pdf]

5. user (2026-06-16T03:23:21.638Z)

Web search results for query: "timestamp supervision temporal action segmentation comparable to fully supervised SF-Net Li 2021"

Links: [{"title": "A Generalized and Robust Framework for Timestamp Supervision in Temporal Action Segmentation | SpringerLink", "url": "https://link.springer.com/chapter/10.1007/978-3-031-19772-7_17"}, {"title": "[2103.06669] Temporal Action Segmentation from Timestamp Supervision", "url": "https://arxiv.org/abs/2103.06669"}, {"title": "Temporal Action Segmentation: A Comprehensive Overview", "url": "https://www.emergentmind.com/topics/temporal-action-segmentation"}, {"title": "CVPR 2021 Open Access Repository", "url": "https://openaccess.thecvf.com/content/CVPR2021/html/Li_Temporal_Action_Segmentation_From_Timestamp_Supervision_CVPR_2021_paper.html"}, {"title": "Temporal Action Segmentation from Timestamp Supervision | IEEE Conference Publication | IEEE Xplore", "url": "https://ieeexplore.ieee.org/document/9577408/"}, {"title": "Temporal Action Segmentation from Timestamp Supervision", "url": "https://github.com/ZheLi2020/TimestampActionSeg"}, {"title": "Temporal Action Segmentation from Timestamp Supervision | Request PDF", "url": "https://www.researchgate.net/publication/350005449_Temporal_Action_Segmentation_from_Timestamp_Supervision"}]

Based on the search results, here's what I found regarding timestamp supervision in temporal action segmentation compared to fully supervised approaches and SF-Net:

Key Findings

Li et al. (2021) proposed using timestamp supervision for temporal action segmentation, where timestamps require a comparable annotation effort to weakly supervised approaches while providing more supervisory signal.

Performance Compared to Fully Supervised Methods

Models trained with timestamp annotations achieve comparable performance to fully supervised approaches. More specifically, state-of-the-art timestamp supervision halves the gap to full supervision, with an accuracy of 64.1% vs. 68.0% on the Breakfast dataset.

How It Works

The approach uses model output and annotated timestamps to generate frame-wise labels by detecting action changes. The method introduces a confidence loss that forces predicted probabilities to monotonically decrease as the distance to the timestamps increases.

Relationship to SF-Net

SF-Net is described as a single-frame supervision approach for temporal action localization, and both SF-Net (Ma et al., 2020) and the Li et al. (2021) timestamp supervision approach are referenced together in the literature as related weak supervision methods for temporal action tasks.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-06-16T03:23:24.978Z)

The PDF didn't parse. Let me get the readable version of the primary source (arXiv 2210.10352) via the abstract page and verify the claim's specifics. The search already gave a critical data point: timestamp supervision "halves the gap" rather than fully closing it (64.1% vs 68.0% on Breakfast).

7. user (2026-06-16T03:23:30.334Z)

Based on the web page content provided, I cannot supply the specific accuracy numbers you're requesting. The page only contains the abstract and metadata for the paper arXiv:2103.06669, which does not include detailed numerical results.

The abstract does make a performance claim: "models trained with timestamps annotations achieve comparable performance to the fully supervised approaches." However, the exact accuracy percentages for Breakfast, GTEA, and 50Salads datasets along with any remaining performance gaps are not presented in this excerpt.

To obtain those specific numbers, you would need to access the full PDF of the paper itself, which is available via the link provided on the arXiv page.

8. user (2026-06-16T03:23:30.395Z)

Paper Overview

Title: Temporal Action Segmentation: An Analysis of Modern Techniques

Authors: Guodong Ding, Fadime Sener, Angela Yao

Venue: TPAMI 2023 (IEEE Transactions on Pattern Analysis and Machine Intelligence)

Topic: Computer Vision and Pattern Recognition

Survey Status

Yes, this is a comprehensive survey. The abstract states: "This survey analyzes and summarizes the most significant contributions and trends" in temporal action segmentation, examining "task definition, common benchmarks, types of supervision, and prevalent evaluation measures."

Timestamp Supervision Findings

The provided content does not include specific details about timestamp supervision performance or comparisons with fully supervised approaches. While the abstract mentions the paper "systematically investigate[s]" essential techniques and conducts "a thorough review of existing TAS works categorized by their levels of supervision," the actual findings on this topic are not available in the webpage excerpt provided.

To find this information, you would need to access the full PDF of the paper.

9. assistant (2026-06-16T03:23:35.880Z)

I now have key facts. The primary source (arXiv 2210.10352) is a legitimate, well-regarded TPAMI 2023 survey "Temporal Action Segmentation: An Analysis of Modern Techniques" by Ding, Sener, Yao. The claim's quote describes timestamp methods performing "comparably with fully supervised approaches" ? which matches the underlying primary literature (Li et al. CVPR 2021, arXiv 2103.06669), whose own abstract states "models trained with timestamps annotations achieve comparable performance to the fully supervised approaches."

Let me probe for the most important refutation angle: is "comparable" an overreach? The earlier search showed timestamp supervision only "halves the gap" (64.1% vs 68.0% on Breakfast). Let me also check whether there's contradicting evidence about robustness/qualification, and verify the exact survey wording.

10. user (2026-06-16T03:23:41.922Z)

Web search results for query: "Ding Sener Yao "Temporal Action Segmentation" survey timestamp supervision "comparable" fully supervised"

Links: [{"title": "[2103.06669] Temporal Action Segmentation from Timestamp Supervision", "url": "https://arxiv.org/abs/2103.06669"}, {"title": "Temporal Action Segmentation from Timestamp Supervision - ADS", "url": "https://ui.adsabs.harvard.edu/abs/2021arXiv210306669L/abstract"}, {"title": "A Generalized and Robust Framework for Timestamp Supervision in Temporal Action Segmentation | SpringerLink", "url": "https://link.springer.com/chapter/10.1007/978-3-031-19772-7_17"}, {"title": "Temporal Action Segmentation from Timestamp Supervision", "url": "https://arxiv.org/pdf/2103.06669"}, {"title": "Robust Action Segmentation from Timestamp Supervision", "url": "https://arxiv.org/pdf/2210.06501"}, {"title": "A Generalized & Robust Framework For Timestamp Supervision in Temporal Action Segmentation", "url": "https://arxiv.org/pdf/2207.10137"}, {"title": "Unified Fully and Timestamp Supervised Temporal Action Segmentation via Sequence to Sequence Translation | Springer Nature Link", "url": "https://link.springer.com/chapter/10.1007/978-3-031-19833-5_4"}, {"title": "SMC-NCA: Semantic-guided Multi-level Contrast for Semi-supervised Temporal Action Segmentation", "url": "https://arxiv.org/pdf/2312.12347"}, {"title": "[2210.10352] Temporal Action Segmentation: An Analysis of Modern Techniques", "url": "https://www.arxiv-vanity.com/papers/2210.10352/"}]

I found the research you're looking for. The survey on Temporal Action Segmentation was authored by Guodong Ding and Angela Yao from the National University of Singapore, with Fadime Sener from Meta Reality Labs.

The key finding you mentioned is clearly stated: "Timestamps require a comparable annotation effort to weakly supervised approaches, and yet provide a more supervisory signal." This indicates that timestamp supervision represents a middle ground between weakly supervised and fully supervised approaches in terms of both annotation cost and supervision quality.

The paper proposes using timestamp supervision for the temporal action segmentation task, and research has shown that timestamps for training reported competitive results compared to fully supervised learning.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

11. user (2026-06-16T03:23:45.306Z)

Web search results for query: "timestamp supervision" temporal action segmentation limitations annotation noise robustness gap to full supervision 2022 2023"

Links: [{"title": "A Generalized and Robust Framework for Timestamp Supervision in Temporal Action Segmentation | SpringerLink", "url": "https://link.springer.com/chapter/10.1007/978-3-031-19772-7_17"}, {"title": "Temporal Action Segmentation from Timestamp Supervision - R Discovery", "url": "https://discovery.researcher.life/article/temporal-action-segmentation-from-timestamp-supervision/c01ca183a2cb3231aeac995d5174d093"}, {"title": "A Generalized & Robust Framework For Timestamp Supervision in Temporal Action Segmentation", "url": "https://arxiv.org/pdf/2207.10137"}, {"title": "Temporal Action Segmentation from Timestamp Supervision", "url": "https://arxiv.org/pdf/2103.06669"}, {"title": "[2207.10137] A Generalized & Robust Framework For Timestamp Supervision in Temporal Action Segmentation", "url": "https://arxiv.org/abs/2207.10137"}, {"title": "A Generalized & Robust Framework For Timestamp Supervision in", "url": "https://www.ecva.net/papers/eccv_2022/papers_ECCV/papers/136640276.pdf"}, {"title": "Temporal Action Segmentation: Weak Supervision and Efficiency", "url": "https://bonndoc.ulb.uni-bonn.de/xmlui/handle/20.500.11811/11007"}, {"title": "Robust Action Segmentation from Timestamp Supervision", "url": "https://arxiv.org/pdf/2210.06501"}, {"title": "Timestamp-supervised Wearable-based Activity Segmentation and Recognition with Contrastive Learning and Order-Preserving Optimal Transport", "url": "https://arxiv.org/pdf/2310.09114"}, {"title": "Temporal Action Segmentation: A Comprehensive Overview", "url": "https://www.emergentmind.com/topics/temporal-action-segmentation"}]

Based on the search results, here's what I found regarding timestamp supervision in temporal action segmentation, particularly its limitations, annotation noise robustness, and the gap to full supervision:

Key Findings

****Core Limitations of Timestamp Supervision:****

Previous works on timestamp supervision rely on assigning hard labels to unlabelled frames, and performance rapidly collapses under subtle violations of the annotation assumptions. Additionally, annotators' slow performance and mistakes highlight the impracticality of the constraints of Timestamp Supervision.

****Annotation Noise and Robustness Issues:****

Existing methods generate pseudo frame-wise labels iteratively based on model outputs and timestamp annotations, but this practice may introduce noise and oscillation during training, leading to performance degeneration. Previous works assume that every action segment is annotated with a timestamp which is an unrealistic and restrictive assumption, and robustness against missing annotations is of crucial importance as such mistakes can occur during the annotation process.

****Gap to Full Supervision:****

State-of-the-art timestamp supervision halves the gap to full supervision (Acc=64.1% vs. 68.0% on Breakfast).

****Recent Solutions (2022-2023):****

To address these challenges, a novel Expectation-Maximization (EM) based approach leverages the label uncertainty of unlabelled frames and is robust enough to accommodate possible annotation errors. Additionally, a teacher model parallel to the segmentation model helps stabilize the process of model optimization and suppress noise to improve the stability of pseudo labels.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

12. assistant (2026-06-16T03:23:58.860Z)

I now have enough evidence to render a verdict. Let me assess the claim against the checklist.

****Checklist assessment:****

1. ****Supported by the quote?**** Yes. The claim states timestamp supervision "achieves performance comparable to fully supervised methods, making it a compelling low-label direction." The supporting quote from the survey says verbatim: "timestamp methods perform comparably with fully supervised approaches, making it a compelling direction for further exploration." This is a faithful paraphrase ? the claim mirrors the quote almost word-for-word, including the "compelling direction" framing. No overreach in the paraphrase itself.
2. ****Contradicting evidence?**** The underlying claim "comparable to fully supervised" is the **standard, well-established** finding in this subfield, originating with Li et al. (CVPR 2021, arXiv 2103.06669), whose own abstract states: "models trained with timestamps annotations achieve comparable performance to the fully supervised approaches." The survey is reporting an accurately characterized community consensus, not a fringe claim. Nuance exists ? "halves the gap" (64.1% vs 68.0% on Breakfast = a real ~4-point residual gap), and follow-up work (arXiv 2207.10137, 2210.06501) shows performance can "collapse under subtle violations of annotation assumptions." But these are **qualifications**, not contradictions: "comparable" (not "equal") accurately describes a small residual gap, and the robustness concerns motivate refinements that **preserve** the comparable-performance result. None of this refutes the claim as stated.
3. ****Source quality vs claim strength?**** Excellent. Primary source is a TPAMI 2023 survey by Ding, Sener & Yao ? a top-tier venue, authoritative survey, exactly the right source type. The core empirical claim traces to a CVPR 2021 primary paper. This is not marketing, press release, forum speculation, or a cherry-picked single benchmark.
4. ****Outdated?**** No. 2021?2023 work; the survey is 2023. Timestamp supervision remains an active, recognized low-label TAL direction. Fully current for a June 2026 report.
5. ****Marketing/cherry-pick?**** No. Peer-reviewed survey + CVPR primary paper.

The claim is well-supported, current, and from an authoritative source. The only honest caveat (which the report should carry, but which does NOT refute the claim) is that "comparable" means a small residual gap remains and robustness depends on annotation-assumption validity. The claim does not assert "equals" or "no gap," so it survives.

13. user (2026-06-16T03:24:10.734Z)

Structured output provided successfully

14. assistant (2026-06-16T03:24:15.389Z)

Verdict: ****refuted=false**** (claim survives). The claim is a faithful paraphrase of the supporting quote, sourced from an authoritative TPAMI 2023 survey (Ding, Sener & Yao) whose core finding traces to the seminal CVPR 2021 primary paper (Li et al., arXiv:2103.06669), which states verbatim that timestamp-trained models "achieve comparable performance to the fully supervised approaches." It is current, peer-reviewed, and not marketing. The only honest caveat ? which qualifies rather than refutes ? is that "comparable" means a small residual gap (~64.1% vs 68.0% on Breakfast) and that performance is sensitive to annotation-assumption violations, so the report should carry that nuance plus the note that the benchmarks are general action-segmentation datasets, making transfer to the 1-D shuttle setting a reasonable but unverified extrapolation.